

Jeremy D. Forsythe

EDUCATION

Clemson University (Ph.D. Forestry & Environmental Conservation - May 2026) Clemson, SC

- **Award:** Phi Kappa Phi Honor Society Membership
- **Coursework:** Applied Data Science & Machine Learning, Statistical Methods, Global Change Ecology, Micrometeorology, Quantitative Ecology, Scientific Writing, Wetland Biology, Ecology of Freshwater Forests, Fire Ecology & Management

University of Kansas (M.A. Ecology & Evolutionary Biology - Aug 2018) Lawrence, KS

- **Award:** Department of Ecology and Evolutionary Biology Summer Fellowship
- **Coursework:** Biometry, Forest Ecosystems, Simulation Modeling, Plant Ecology, Likelihood Methods, Plant Systematics, Responsible Research & Teaching Effectiveness

University at Albany (B.S. Biological Sciences - Dec 2013) Albany, NY

- **Award:** Dean's List
- **Coursework:** Theoretical Ecology, Experimental Ecology, Biological Consequences of Climate Change, Geographic Information Systems, Statistical Inference, Evolution, Behavioral Ecology, Cell Biology, Genetics, Grazing in Terrestrial Ecosystems (Graduate Level), Integrated Principles of Ecology (Graduate Level)

Hudson Valley and Broome Community Colleges Troy/Binghamton, NY

- **Award:** Dean's List
- **Coursework:** Field Ecology, Zoology, Animal Behavior, Organic Chemistry, Introductory Science Courses, Multi-Variate Calculus, Differential Eq.

SKILLS

- Acquired substantial skills in satellite remote sensing, eddy covariance, & field ecology methods through experience in research & coursework, including flux sensor installation & maintenance, data processing (i.e. fluxes, partitioning, & gapfilling), LAI measurements, FIA style forest surveys, plant identification, GIS, and remote sensing (satellite and ground based).
- Strong working knowledge of relevant software: *R*, *Python*, *LaTeX*, *Eddypro*, *REddyProc*, *caret* & *Scikit-learn*, *Google Earth Engine*, *ArcGIS/QGIS*, *Microsoft Office*.
- Practical skills developed in and out of academia include excellent public speaking, writing, and communication, ability to work as part of a team, leadership, dependable work ethic, time management, a clean driver's license, ability to operate manual transmission vehicles, an above average amount of patience, and an eccentric sense of humor.

PROFESSIONAL EXPERIENCE

Research Associate, Northern Arizona University & Bonanza Creek LTER 2024–Present

- Liaison between the Bonanza Creek Long-Term Ecological Research Program and NASA's Arctic-Boreal Vulnerability Experiment (ABoVE), bridging the Mack/Walker lab (field ecology, wildfire) and the Goetz lab (satellite remote sensing).
- Lead researcher quantifying how decreasing fire return intervals reshape boreal forest structure, fuels, and carbon balance across Alaska and western Canada using harmonized fire perimeters, Landsat burn severity, and climate reanalysis data.
- Contributed to a meta-analysis of North American Arctic and boreal aboveground biomass datasets.

Independent Contractor, Chapman University & NASA ECOSTRESS 2022–2024

- Co-developed and instructed the ICE CREAM (Integrating Communication of ECOSTRESS into Community Research, Education, Applications, and Media) open-access satellite remote sensing curriculum with Drs. Joshua Fisher and Gregory Goldsmith.

- Developed course tutorials, delivered lectures, and mentored students using NASA AppEEARS and QGIS.
- Curriculum adopted by 12+ minority-serving institutions; accessed by 1,400+ users.
- Independent assessment showed significant gains in science interest (20%), belonging (11%), science identity (9%), and content understanding (19%) across diverse student populations.

TEACHING & OUTREACH

Northern Arizona University

- Instructor of Record: Independent study adaptation of *Observing Earth from Above* satellite remote sensing course (Spring 2025).
- Guest Lecturer: Machine learning for ecologists and regression methods in ecological data analysis courses (Dr. Xanthe Walker).
- Leading a Machine Learning for Ecologists workshop at the 2026 Ecological Society of America meeting.
- Founded “Snack Overflow,” an inclusive coding and writing working group for undergraduates and graduate students.
- Presented research biannually at the Alaska Fire Science Consortium.
- Represented NASA ABoVE and NSF at the Interagency Arctic Research Policy Committee (IARPC) meeting connecting tribal leaders with researchers.

Clemson University

- Mentored through Clemson’s UPIC Program, advising 1-2 undergraduate students each summer through a research program and presentation of an independent project.
- Data Management and Visualization in Natural Resources - Designed and led a guest lecture for a graduate level class teaching techniques on how to organize, manipulate, and visualize biological data in the R programming language.
- Guest Lecturer: Invasive plant species ecology in a wetland biology course.
- Founded “Snack Overflow” coding and writing working group (later replicated at NAU).
- Active member of Ameriflux DEI & FLUXNET Education Committees

University of Kansas

- Teaching Assistant (2 semesters): BIO 152 Principles of Biology. Collaborated on flipped classroom redesign with Dr. Mark Mort; authored exams, lectured, and ran lab sections.
- Assisted in developing and executing a course redesign for an undergraduate level introductory course for biology majors that emphasized alternative learning strategies such as learning groups, real time response surveys, and integrated on-line tools.
- Mentored through the NSF’s REU program, instructing a cohort of 12 students through a 10 week research program that included the design, implementation, and presentation of an independent experiment.

RESEARCH EXPERIENCE

Northern Arizona University Research Associate

- Quantified the spatial extent of boreal reburns and assessed how fire weather, land cover, and fire behavior change across fire return intervals using harmonized fire perimeters (1940–2023), Landsat burn severity, and ERA5 climate reanalysis across Alaska and western Canada.
- Found that ~20% of area burned since 1985 was reburned within 70 years; demonstrated that high vapor pressure deficit amplifies fire size and severity in young stands, partially decoupling fire behavior from fuel limitation.
- Contributed to meta-analysis of North American Arctic and boreal aboveground biomass datasets assessing accuracy and dynamics of satellite-based carbon monitoring.

Chapman University / NASA ECOSTRESS Independent Contractor

- Developed open-access satellite remote sensing tutorials for the ICE CREAM curriculum using NASA ECOSTRESS data, AppEEARS, and QGIS.

Clemson University Ph.D.

- Dissertation
 - Examined the drivers of photosynthetic uptake of atmospheric carbon by coastal ecosystems, including diffuse radiation enhancement, an effect of the light scattering from clouds and aerosols that often increases photosynthesis by more evenly distributing the incoming solar energy through the canopy.
 - Developed ecosystem management support tools for coastal ecosystems and Southern pine forests by combining flux tower and satellite remote sensing vegetation indices to monitor long-term changes in carbon sequestration with a newly optimized “two-leaf” light use efficiency (LUE) model.
 - Improved understanding of the physical mechanisms controlling the coupling of evapotranspiration to carbon fluxes in coastal Southern pine forests by investigating the effects of light quality in addition to quantity on water cycling and productivity.
- Labwork
 - Worked with my lab and collaborators to build a new eddy covariance mesonet for measuring greenhouse gas fluxes in coastal South Carolina, including tower construction, sensor calibration & maintenance, and data workflow from raw measurements through a final partitioned & gapfilled product (i.e. QA/QC, developing machine learning algorithms, and an Ameriflux submission pipeline).

University of Kansas Master’s Degree

- Joined a long term community analysis project exploring forest succession in an ecotone between eastern deciduous forests and tall-grass prairies.
- Adapted a novel statistical methodology for interpreting the underlying distribution of diameters for tree populations using maximum likelihood.

University of Kansas National Science Foundation Research Experience for Undergraduates

- Developed an independent research project on the interactive effects of management on the restoration of native tall grass prairie across increasing spatial scales.

University at Albany

- Developed an invasive species management project using ungulate grazing as a control.

REFERENCES

- Dr. Michelle Mack: Research Associate Advisor; Regents Professor at Northern Arizona University.
michelle.mack@nau.edu
- Dr. Xanthe Walker: Research Associate Advisor; Assistant Professor at Northern Arizona University.
xanthe.walker@nau.edu
- Dr. Scott Goetz: Research Associate Advisor; Regents Professor at Northern Arizona University.
scott.goetz@nau.edu
- Dr. Joshua Fisher: Mentor; Presidential Fellow of Ecosystem Science at Chapman University.
jbfisher@chapman.edu
- Dr. Tom O’Halloran: Ph.D. Advisor; Associate Professor at Clemson University.
tohallo@clemson.edu (843) 546-1013

- Dr. Bryan Foster: Master's Advisor, REU Mentor, and University of Kansas Faculty. bfoster@ku.edu (785) 864-4361

PUBLICATIONS

- {Under review at Fire Ecology} Forsythe, J.D., Mack, M., Berner, L., Goetz, S., Hansen, W., Hayes, K., Johnstone, J., Potter, S., Rogers, B., Talucci, A., Wang, J., & Walker, X. Short-Interval Wildfires Are Smaller, Less Severe, and Require More Extreme Weather than Long-Interval Fires in Boreal North America *Fire Ecology*.
- {Under review at JGR Biogeosciences} Forsythe, J.D. & O'Halloran, T.L. Light quality controls daily LUE in Southern pine forests: an interpretable machine learning approach. *JGR Biogeosciences*.
- {In Preparation} Forsythe, J.D., O'Halloran, T.L., & Hawman, P. Satellite data infused light use efficiency models reveal importance of diffuse light fertilization in U.S. Southern pine ecosystems. *Agricultural and Forest Meteorology*.
- {In Preparation} Forsythe, J.D., O'Halloran, T.L., Thomas, R.Q., Fisher, J.B., & Duberstein, J. Diffuse light increases evapotranspiration and water use efficiency in Southern pine. *Ecohydrology*.
- Liang, W., Bauck, K., Maloney, D., et al. including Forsythe, J. (2026). Meta-analysis of North American Arctic and boreal aboveground biomass datasets. *Environmental Research Letters*, 21, 054004.
- Zhu, Q., Chen, J., Bourque, C.P.A., et al. including Forsythe, J.D. (2024). Albedo-induced global warming potential following disturbances in global temperate and boreal forests. *JGR Biogeosciences*, 129.
- Aguilos, M., Sun, G., Liu, N., et al. including Forsythe, J. (2024). Energy availability and leaf area dominate control of ecosystem evapotranspiration in the southeastern U.S. *Agricultural and Forest Meteorology*, 349.
- Williams, T.M., B. Williams, B. Song, T.L. O'Halloran, J.D. Forsythe. 2022. Mapping natural forest stands with low-cost drones. *Math. and Comp. Forestry & Nat.-Res. Sci.*, 14(1), 22–42.
- Ahlswede, B.J., O'Halloran, T.L., Forsythe, J.D., & Thomas, R.Q. 2021. A minimally managed switchgrass ecosystem in a humid subtropical climate is a source of carbon to the atmosphere. *GCB Bioenergy*, 14, 24–36.
- Williams, T.M., T.L. O'Halloran, B. Song, J.D. Forsythe, and B.J. Williams. 2021. Sources of error from dense understory vegetation in Coastal Plain forest hydrologic analyses using LiDAR DEMs. *In Proc. 13th Southern Forestry and Natural Resource Management GIS Conf.* Athens, GA.
- Williams, T.M., T.L. O'Halloran, B. Song, J.D. Forsythe, and B.J. Williams. 2020. Evaluating high water table hydrology and eddy covariance measurements of evapotranspiration at a newly instrumented watershed in coastal South Carolina. *In Proc. 7th Interagency Conf. on Research in the Watersheds*. Tifton, GA.
- Forsythe, J.D., O'Halloran, T.L., & Kline, M.A. 2020. An eddy covariance mesonet for measuring greenhouse gas fluxes in coastal South Carolina. *Data*, 5, 1–20.

PRESENTATIONS

- **Talk:** Forsythe J, Potter S, Rogers B, Berner L, Goetz S, Johnstone J, Mack M, Walker X. Quantifying The Controls On Boreal Forest Reburning With Interpretable Machine Learning. American Geophysical Union (AGU); 2025.
- **Talk:** Hall J, Hayes K, Asena Q, Forsythe J, et al. Predicting forest burn risk from 2026 to 2040 in interior Alaska. American Geophysical Union (AGU); 2025.
- **Talk:** Liang W, Bauck K, Maloney D, et al. including Forsythe J. Meta-analysis of North American Arctic and Boreal Aboveground Carbon Maps to Provide Guidance for User Communities. American Geophysical Union (AGU); 2025.
- **Talk:** Forsythe J, Potter S, Rogers B, Berner L, Goetz S, Johnstone J, Mack M, Walker X. Legacy of Wildfire in Boreal Forests: Characterizing Fire-on-Fire Interactions with Satellite Remote Sensing. Bonanza Creek LTER All Hands Meeting; 06/2025.

- **Talk:** Forsythe J, Rogers B, Berner L, Goetz S, Johnstone J, Mack M, Walker X. Drivers and Impacts of Reburning in Boreal Forest Ecosystems (DIRE). NASA ABoVE Team Meeting; 05/2025.
- **Poster:** Maloney D, Hoy E, Larson E, Montesano P, Wang J, Liang W, Bauck K, Walker X, Forsythe J. A Comparative Analysis of Available Arctic and Boreal Biomass Datasets. NASA ABoVE Team Meeting; 05/2025.
- **Invited Talk:** Forsythe J, Mack M, Walker X. Decades of Discovery: How NSF's Long Term Ecological Research is Advancing Arctic Understanding. Interagency Arctic Research Policy Committee (IARPC); 2025.
- **Invited Talk:** Forsythe J, Potter S, Rogers B, Berner L, Goetz S, Johnstone J, Mack M, Walker X. Mapping Top-Down and Bottom-Up Controls on Wildfire in Boreal Forest Ecosystems. Interagency Arctic Research Policy Committee (IARPC); 2025.
- **Talk:** Forsythe J, Potter S, Rogers B, Berner L, Goetz S, Johnstone J, Mack M, Walker X. Top-Down and Bottom-Up Controls on Short Interval Wildfire in Boreal Forest Ecosystems. American Geophysical Union (AGU); 2024.
- **Invited Talk:** Forsythe J. Pixels, Patterns, and Plants: Integrating Remote Sensing, Micrometeorology, and Field Ecology to Unravel Ecosystem Functioning Across Diverse Landscapes. NAU Ecoinformatics Seminar Series; 2024.
- **Poster:** Forsythe J, Potter S, Rogers B, Berner L, Goetz S, Johnstone J, Mack M, Walker X. Evidence of Reburning in Boreal Forest Ecosystems from Landsat Observations. NASA ABoVE Team Meeting; 05/2024.
- **Invited Talk:** Forsythe JD, O'Halloran TL. Diffuse Light & Evapotranspiration in Southern Pine Forests. NASA ECOSTRESS Science and Application Team Meeting; Fall 2023.
- **Invited Talk:** Forsythe JD. Writing Tutorials For Accessing & Visualizing ECOSTRESS Data With Open Source Tools. NASA ECOSTRESS Science and Application Team Meeting; Fall 2023.
- **Talk:** Forsythe JD, O'Halloran TL. Responses of Southern Pine Forest Gross Primary Productivity to Diffuse Radiation: A 17 Site-Year Study. Ecological Society of America; 08/2023.
- **Talk:** Forsythe JD, O'Halloran TL. Sensitivity of Diffuse Light Fertilization in Southern Pine Carbon Fluxes Observed with Remotely-Sensed Vegetation Indices. American Meteorological Society 35th Conference; 05/2023.
- **Poster:** Forsythe JD, O'Halloran TL, Wise M, Balkcum O, Song B, Kline M A, DeGarady C. [How do fire & management alter longleaf pine understory carbon stocks & whole ecosystem carbon sequestration?](#) 14th Biennial Longleaf Conference; 10/2022.
- **Invited Talk:** Carbon Management & Clearcut Recovery in Longleaf Pine Ecosystems. Forsythe JD, O'Halloran TL. Jones Center at Ichauway. Newton, GA. 4/2022.
- **Invited Talk:** Managing land to sequester carbon and greenhouse gases. O'Halloran, TL., JD Forsythe, L. Clay. SC Soil & Water Conservation Society. Charleston. 2/2022.
- **Invited Talk:** Carbon and climate benefits of longleaf pine forests. O'Halloran, T.L., C. DeGarady, M. Motallebi, Song, BS., L. Clay, JD Forsythe. SC Native Plant Society, Lowcountry Chapter (virtual). 2/2022.
- **Poster:** Marshes Migrating Into Forests: Effects of Sea Level Rise on Coastal Ecosystems. Balkcum O, Song B, Forsythe JD, Wise M, Kline MA, O'Halloran TL. Clemson University Baruch Institute of Coastal Ecology and Forest Science Behind the Gate; Georgetown SC, 7/2021.
- **Poster:** How Does Fire Alter Carbon Stocks and Plant Diversity in a Longleaf Pine Understory? Wise M., Forsythe JD, Song B., Balkcum O., Kline MA, O'Halloran TL. Clemson University Baruch Institute of Coastal Ecology and Forest Science Behind the Gate; Georgetown SC, 7/2021.
- **Poster:** Role of Phenology in Switchgrass Carbon Fluxes. Kuleba A, O'Halloran TL, Rady J, Powell L, Forsythe JD. Clemson University Baruch Institute of Coastal Ecology and Forest Science Behind the Gate; Georgetown SC, 7/2021.

- **Poster:** Comparing Carbon Dioxide and Methane Fluxes between a Natural Salt Marsh and Managed Wetlands. Clay, L., Forsythe, JD, and O'Halloran, TL Behind the Gate. Baruch Institute for Coastal Ecology and Forest Science, 7/2021.
- **Poster:** Forsythe JD, O'Halloran TL, Williams T, Kaminski R, Kline MA. [An Eddy Covariance Mesonet Measuring Coastal Carbon Fluxes in South Carolina](#). 7th North American Carbon Program OSM; 3/2021.
- **Poster:** Forsythe, JD, O'Halloran, TL and Kline, MA. [Establishing An Eddy Covariance Mesonet in Coastal South Carolina](#) Ameriflux; 2020.
- **Poster:** Forsythe JD, Foster, BL. [The Effects Of Disturbance And Soil Nutrient Enrichment On Grassland Community Biodiversity Across Spatial Scales](#). KU Madison & Lila Self Graduate Fellowship Symposium